

PROFILE

Nati is a motivated electrical engineer with software and hardware experience and an understanding of PC system architecture. He learns quickly, adapts well in fast-paced environments, and works effectively with stakeholders to meet tight deadlines. Passionate about solving problems, he is eager to gain hands-on experience and contribute to innovative and impactful projects.

EXPERIENCE

Dell Inc., Austin, TX

Hardware Engineering Rotation Program

February 2023 – Present

- Option module complexity reduction by investigating feasibility on a USB to serial conversion to remove \$0.13 cost adder to high volume platform (Dell Optiplex, 10M+ units per year)
- 2 patents pending for Secure USB IP (hardware-based USB traffic monitoring for customers with concerns of security over USB) . Independently designed and developed a PCB proof of concept to validate Secure USB IP (Idea to gerberout).
- Traveled abroad to Mexico and Taiwan for hands-on lab training and factory visits to observe high-volume consumer electronics manufacturing.
- Built a custom Dust chamber to investigate high dust related recall issues on Inspiron devices in India.
- architect a low-power contextual sensing system using an MCU with an integrated NPU and multiple sensors to run AI models for detecting device context (in bag, on desk, user presence, etc.).
- Developed an evaluation Dev kit for the sensing architecture to support early SW/FW development—responsible for architecture and Schematic.
- Worked on evaluating self capacitance sensing technology as a human touch and proximity sensor using the laptop metal housing as the sensing node(needs floating housing to work). make multiple POC iterations to improve proximity performance (2-3 ft detection).
- Led testing efforts to evaluate and mitigate the negative effects of a floating housing, including RF desense, EMI, and ESD.
- Conducting competitive analysis. automate benchmark running, data collection and reporting with scripts
- Schematic capture for ARL based Notebook, responsible for HDMI, EC and DC-DC converters
- Wrote an automation script for netlist comparisons. Helped save time with redrawing schematics for readability.
- Leveraged AI to make a full stack website that helps with competitive analysis.

Undergraduate Systems Architect Intern

June 2022 - August 2022

- Helped develop a standard module connector pinout, addressing high-speed PCB issues such as crosstalk, impedance matching, shielding, and connector cost tradeoffs.

Undergraduate Electrical engineer Intern

May 2021 - August 2021

- Routed an SLI bridge for the Nvidia GV100 card that would be utilized on a Dell Precision tower.

Undergraduate BIOS Intern

June 2020 - July 2020

- Implemented Bluetooth 4.0 in BIOS by integrating a Bluetooth driver to enable Bluetooth device connection on pre-boot

Nokia, Irving, TX

5GRAN E2E Testing and Automation Intern

June 2019 - August 2019

- Built a Python GUI to automate Excel-to-JIRA data entry, saving ~1.5 hours of manual work per use.

EDUCATION

Bachelor of Science - BS, Computer Engineering

December 2022

The University of Texas at Dallas

SKILLS

System Architecture, Cadence/Allegro, KiCad, basic soldering/rework, lab equipment, Vendor engagement, microsoft office, C/C++, Python